DECK 01 · WEEK 1 · MORNING 1

Time Value of Money & DCF

The financial-math foundation every deck that follows will draw from

CFA Level I · Vol 1 Learning Modules 1–2

Roadmap — Time Value of Money & DCF Foundations

6 sections · ~2 hours · Built bottom-up from CFA readings

01	**What is an interest rate?**	3 interpretations · decomposing r · risk premia · CFA Example 1 five bonds · nominal vs real
02	**Measuring return**	HPR · arithmetic / geometric / harmonic means · trimmed-winsorized · money- vs time-weighted
03	**Annualizing returns**	periodic → annual · EAR · continuous compounding · gross / net / real / levered views
04	**Future value**	single CF · compounding frequency · FV of an annuity (retirement savings)
05	**Present value**	single CF · annuity · perpetuity · bond pricing (India, Greece) · Gordon DDM (Shipline)
06	**NPV, IRR & applications**	NPV decision rule · when IRR breaks · mortgage amortization · TVM archetype map

Learning objectives

By the end of this deck you should be able to:

1. Read an interest rate three ways — required return, discount rate, opportunity cost — and back out each risk premium from a set of traded bonds.

2. Choose the correct mean (arithmetic, geometric, harmonic, trimmed/winsorized) for a given return-aggregation problem and justify the choice.

3. Distinguish money-weighted from time-weighted returns and explain why the latter is the GIPS standard for evaluating managers.

4. Convert any nominal rate into an EAR or a continuously compounded rate and move a single cash flow anywhere on a timeline.

5. Price a plain-vanilla bond, value a stock with a constant- or multi-stage DDM, and evaluate a capital project using NPV (and know when IRR misleads).

SECTION 01

What is an interest rate?

An interest rate is simultaneously a required return, a discount rate, and an opportunity cost — three views of the same number

If USD 9,500 today and USD 10,000 in one year are equivalent in value, then r = 500 / 9,500 = 5.26%.

$$r = (FV - PV) / PV \rightarrow (10{,}000 - 9{,}500) / 9{,}500 = 5.26\%$$

1 **Required rate of return**
The MINIMUM rate an investor must receive to accept an investment. Below this rate, they walk away.

2 **Discount rate**
The rate used to translate a future value back to present — the inverse operation of compounding.

3 **Opportunity cost**
What is forgone by choosing a course of action. Spending USD 9,500 today forgoes 5.26% earned by saving.

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, pp. 4–7.

Every market interest rate decomposes into a real risk-free rate plus four risk premia

The additive shortcut is fine for quick mental math; compounding is exact. Indicative ranges are post-2000 developed-market averages.

$$r = r_{real\text{-}rf} + \text{Inflation} + \text{Default} + \text{Liquidity} + \text{Maturity}$$

Component	Indicative range	What it compensates for
Real risk-free rate	0.5 – 1.5%	Time preference for current vs future consumption, no inflation, no risk
Inflation premium	2 – 3%	Expected loss of purchasing power over the life of the instrument
Default risk premium	0 – 5%+	Compensation for possible non-payment (issuer credit risk)
Liquidity premium	0.5 – 1.5%	Cost of selling an investment quickly at close to fair value
Maturity premium	0.5 – 2%	Longer bonds are more sensitive to yield moves — an additional risk

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, "Determinants of Interest Rates", pp. 6–8.

CFA LM 1 Example 1 lets us back out each premium by comparing five bonds that differ on one dimension at a time

Scenario — all five investments promise a single payment at maturity. Inflation, liquidity, and default premia are constant across horizons.

Inv.	Maturity	Liquidity	Default	r (%)
1	2y	High	Low	2.0
2	2y	Low	Low	2.5
3	7y	Low	Low	r_3 ?
4	8y	High	Low	4.0
5	8y	Low	High	6.5

SOLUTION — READ OFF THE PREMIA

Compare bonds that differ on exactly one feature to isolate each premium.

1. Liquidity premium: (Inv 2 − Inv 1) = 2.5% − 2.0% = 0.50%
2. Default premium: (Inv 5 − Inv 4) − LP = (6.5% − 4.0%) − 0.50% = 2.00%
3. Upper bound for r_3: Inv 4 adjusted for low liquidity = 4.0% + 0.50% = 4.50%
4. Lower bound for r_3: Inv 2 (same liquidity & default, shorter) = 2.50%

⇒ 2.50% ≤ r_3 ≤ 4.50% (positive maturity premium in between)

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, Example 1, pp. 7–8.

The nominal risk-free rate is the real rate plus expected inflation — exact via Fisher, approximated additively

The approximation drops a 2nd-order term (real × inflation). For developed-market rates it rarely matters; in high-inflation economies it does.

$$(1 + r_{nominal\text{-}rf}) = (1 + r_{real\text{-}rf}) \times (1 + \pi) \approx r_{nominal\text{-}rf} = r_{real\text{-}rf} + \pi$$

where: π = expected inflation over the horizon of the instrument.

UNITED STATES

Short-dated US T-bill yield is the standard nominal r-f proxy for USD-denominated analysis. 90-day T-bill quoted at 3% is already annualized — not the 90-day holding return.

FRANCE

BTF (Bons du Trésor à taux fixe) — French negotiable Treasury bills with maturities up to 1 year. The eurozone risk-free proxy for short horizons.

JAPAN

JGB T-bills issued at 6- and 12-month maturities serve as the JPY short-dated r-f proxy.

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, pp. 6–8.

SECTION 02

Measuring return

A holding-period return captures both income yield and capital gain over any single period

HPR is the single-period building block — every multi-period return measure either averages or compounds HPRs.

$$R = (P_1 - P_0 + I_1) / P_0 = \text{capital-gain yield} + \text{income yield}$$

where: P_0 = beginning price; P_1 = ending price; I_1 = income (dividend, coupon) received at period end.

CFA LM 1 EXAMPLE 2 — SINGLE-QUARTER EQUITY POSITION

Investor buys 100 shares at USD 34.50, receives USD 51.55 dividend, sells at USD 30.50 one quarter later.

1. Cost basis: 100 × 34.50 = USD 3,450
2. Ending value + income: (100 × 30.50) + 51.55 = USD 3,101.55
3. HPR = (3,101.55 − 3,450) / 3,450 = −10.1%
4. Decomposition: dividend yield 51.55 / 3,450 = +1.49%; capital loss −400 / 3,450 = −11.59%; total = −10.1%

HPR = −10.10% (answer C)

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, Example 2, pp. 8–10.

Arithmetic mean return is simple and unbiased for the next-period expected return — but biased up when returns actually compound

The bias is worst when returns alternate between large gains and large losses — precisely the regime where the bias matters most.

$$\bar{R}_i = (R_{i1} + R_{i2} + \ldots + R_{iT}) / T$$

Year	Actual return	Actual balance (EUR)	Balance if earned AM 4%	Error (EUR)
0	—	1.0000	1.0000	0.0000
1	−50%	0.5000	1.0400	+0.5400
2	+35%	0.6750	1.0816	+0.4066
3	+27%	0.8573	1.1249	+0.2676
AM of returns	**4.00%**	**ending 0.8573**	**ending 1.1249**	**bias +31%**

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, pp. 10–12.

Geometric mean return captures actual compound growth of one unit of capital — always ≤ arithmetic, with equality only when all returns are equal

Jensen's inequality: concavity of log(1+x) implies E[log(1+R)] ≤ log(1+E[R]) — which is why GM ≤ AM.

$$\bar{R}_G = [(1 + R_1)(1 + R_2) \ldots (1 + R_T)]^{(1/T)} - 1$$

where: Must compound gross-of-loss factors (1+R) — NOT raw returns. Product of factors must be ≥ 0.

SAME SERIES AS PREVIOUS SLIDE: −50%, +35%, +27%

Compute the geometric mean return and verify it reproduces the actual EUR 0.8573 ending balance.

1. Gross factors: (1 − 0.50) × (1 + 0.35) × (1 + 0.27) = 0.50 × 1.35 × 1.27 = 0.85725
2. Cube root: 0.85725^(1/3) = 0.9500 → $\bar{R}_G$ = −5.00%
3. Sanity check: 1 × $(1 - 0.05)^3$ = 0.857375 ✓(matches actual balance up to rounding)

GM = −5.00% vs AM = +4.00% → GM-AM gap of 9 percentage points driven by the −50% drawdown.

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, pp. 11–12.

Harmonic mean is the right average for prices under a dollar-cost-averaging strategy

Rule of thumb: averaging prices applied to a FIXED amount of cash each period → harmonic. Averaging returns applied to capital → geometric. Averaging quantities → arithmetic.

$$X_H = n / \Sigma_i (1 / X_i) \qquad AM \geq GM \geq HM \quad \text{(equality iff all } X_i \text{ equal)}$$

CFA LM 1 EXAMPLE 7 — PORTFOLIO MANAGER DOLLAR-COST-AVERAGES

PM invests EUR 5,000 every year for 4 years. Share prices observed: EUR 62.00, 76.00, 84.00, 90.00. Average cost per share?

1. Shares bought each year: 5,000/62 = 80.65; 5,000/76 = 65.79; 5,000/84 = 59.52; 5,000/90 = 55.56
2. Total shares: 261.52. Total invested: EUR 20,000.
3. Average cost = 20,000 / 261.52 = EUR 76.48 — which equals the harmonic mean of the four prices.
4. Compare: arithmetic mean = (62+76+84+90)/4 = 78.00 → overstates true average cost.

HM = 76.48 < AM = 78.00. Harmonic avoids the upward bias that would matter at allocation time.

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, Example 7, pp. 13–14.

Trimmed and winsorized means are robust to outliers — crucial when a single extreme observation dominates a small sample

Both measures lower the weight of the tails. Trimmed = delete tails; Winsorized = substitute the boundary value.

TRIMMED MEAN

Drop the top and bottom k% of observations, then take the arithmetic mean of what remains. Common choices: 5% or 10% trim.

Effect: completely removes outliers.

WINSORIZED MEAN

Replace the top k% with the (1−k) percentile and the bottom k% with the k percentile — then arithmetic mean.

Effect: caps outliers at a defensible quantile instead of deleting them.

Measure	Returns kept / replaced	Result (%)
Arithmetic mean	all 10	+5.40
5% trimmed mean	drop top 1 and bottom 1 (8 returns)	+3.10

Money-weighted returns reflect investor cash-flow timing; time-weighted returns isolate manager skill

MWR is an IRR on the portfolio's cash flows. TWR compounds sub-period HPRs, making cash-flow timing irrelevant.

$$\text{TWR:} \quad (1 + R_twr)^n = \Pi\,(1 + R_t) \qquad \text{MWR:} \quad \Sigma\, CF_t / (1 + R_mwr)^t = 0$$

where: R_t = return in sub-period t (between cash flows); CF_t = net cash flow (deposits negative, withdrawals positive).

MONEY-WEIGHTED RETURN (MWR)

An IRR. Weights each sub-period by invested capital, so the RETURN the INVESTOR actually earned given deposits and withdrawals. Right metric for an investor evaluating personal wealth growth.

TIME-WEIGHTED RETURN (TWR)

Compound of sub-period HPRs. Neutral to cash-flow timing. The GIPS-required performance measure for evaluating PORTFOLIO MANAGERS, because deposits and withdrawals are outside their control.

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, pp. 16–20.

Time-weighted return is the preferred performance measure for evaluating portfolio managers

Same portfolio, same dates — TWR and MWR can diverge sharply whenever the investor adds or withdraws cash.

Event	Date	Portfolio value	Cash flow	Sub-period HPR
Initial investment	t = 0	100,000	−100,000	—
End of period 1 (before additional deposit)	t = 1	108,000	0	+8.00%
Investor deposits 50,000 at t = 1	t = 1	158,000	−50,000	—
End of period 2	t = 2	149,100	+149,100	−5.63%

TWR vs MWR

TWR = [(1 + 0.08)(1 − 0.0563)]^(1/2) − 1 = +0.97% · MWR solves the IRR → −1.21%

The manager's sub-periods gained 0.97% p.a. The investor lost 1.21% — the deposit landed right before a drawdown.

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, Worked Examples, pp. 18–20.

SECTION 03

Annualizing returns

Any periodic return can be annualized by compounding — but different compounding frequencies produce different effective annual rates

The stated (nominal) rate is a quoting convention. The EAR is what the investor actually earns in a year.

$$EAR = (1 + r_nominal / m)^m - 1$$

where: r_nominal = quoted annual rate; m = number of compounding periods per year.

Compounding	m	Per-period rate	EAR (r = 6%)	EAR (r = 12%)
Annual	1	6.0000%	6.0000%	12.0000%
Semi-annual	2	3.0000%	6.0900%	12.3600%
Quarterly	4	1.5000%	6.1364%	12.5509%
Monthly	12	0.5000%	6.1678%	12.6825%
Daily	365	0.01644%	6.1831%	12.7475%
Continuous	∞	limit	6.1837%	12.7497%

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, pp. 20–24.

Continuous compounding is the limit as frequency → ∞ and is the standard in derivatives and options models

Continuous rates are additive across time — critical for Black-Scholes and for combining returns over non-integer horizons.

$$FV = PV \times e^{(r \times n)} \qquad r_{cc} = \ln(1 + R_{periodic})$$

where: r_cc = continuously compounded rate; R_periodic = discrete periodic return. r_cc < R_periodic always.

FROM DISCRETE TO CONTINUOUS

A stated annual rate of 12% becomes a continuously compounded rate of ln(1.12) = 11.33%. Lower number — but equivalent economically.

FROM CONTINUOUS TO DISCRETE

A 10% continuously compounded rate corresponds to a discrete annual rate of e^0.10 − 1 = 10.52%. Use one convention consistently inside any model.

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, pp. 24–26.

Gross, net, after-tax, real and leveraged returns are five progressively "adjusted" views of the same underlying asset

Always confirm which of these five a quoted return refers to before comparing assets or managers.

Measure	Definition	Example at 8.00% gross
Gross return	Return before any management / admin fees	8.00%
Net return	Gross minus direct management and admin fees	7.15%
After-tax nominal	Net minus tax on dividends, coupons, realised gains	5.00%
Real return	After-tax nominal deflated by the inflation rate	2.44%
Leveraged return	Return on equity after borrowing (e.g. 50% LTV at 4%)	12.00%

LEVERAGE AMPLIFIES BOTH WAYS

A portfolio returning 8% with 50% borrowed at 4% returns 12% on equity — but a −4% year on the asset becomes −12% on equity. Leverage never lowers risk; it only reshapes outcomes.

Source: CFA Institute, Curriculum Vol 1, Learning Module 1, pp. 26–30.

SECTION 04

Future value

The future value of a single cash flow is PV grown by (1+r) over n periods — the single most-used formula in all of finance

Every other TVM formula (annuity, perpetuity, bond, DDM) is built from this single building block.

$$FV_n = PV \times (1 + r)^n$$

where: PV = present value; r = per-period rate; n = number of periods. Consistency of r and n is non-negotiable.

CASH-FLOW TIMELINE — ALWAYS DRAW ONE

You deposit £10,000 today (t=0) at 5% p.a., compounded annually. What is the balance after 10 years?

1. PV = £10,000; r = 5%; n = 10.
2. FV = 10,000 × (1.05)^10
3. (1.05)^10 = 1.62889
4. FV = £16,289

£16,289 — a 62.9% total gain, or 5.00% p.a.

Source: CFA Institute, Curriculum Vol 1, Learning Module 2, pp. 46–50.

Compounding frequency materially changes the outcome even when the nominal rate is unchanged

£1,000 invested for 5 years at 12% nominal — the gap from annual to continuous is ~3.4% of final value.

$$FV = PV \times (1 + r/m)^{(m \times n)}$$

$$FV_{continuous} = PV \times e^{(r \times n)}$$

Compounding	m	FV after 5y	Uplift vs annual
Annual	1	£1,762.34	–
Semi-annual	2	£1,790.85	+£28.51
Quarterly	4	£1,806.11	+£43.77
Monthly	12	£1,816.70	+£54.36
Daily	365	£1,821.94	+£59.60
Continuous	∞	£1,822.12	+£59.78 (+3.4%)

Source: CFA Institute, Curriculum Vol 1, Learning Module 2, pp. 50–54.

The future value of a level annuity is a closed-form sum of compounded payments — the core formula of retirement planning

Ordinary annuity = payments at period END. Annuity due = payments at period START — multiply ordinary formula by (1+r) to convert.

$$FV_annuity = PMT \times [\ ((1 + r)^n - 1) / r\ ]$$

where: PMT = recurring payment; r = rate per period; n = number of payments.

RETIREMENT-SAVINGS EXAMPLE

You save £10,000 at the end of every year for 20 years in an account paying 6% p.a.

1. PMT = 10,000; r = 6%; n = 20
2. (1.06)^20 = 3.2071
3. ((1.06)^20 − 1) / 0.06 = 2.2071 / 0.06 = 36.7856
4. FV = 10,000 × 36.7856 = £367,856

£367,856 at retirement — of which £200,000 principal and £167,856 compound interest.

Source: CFA Institute, Curriculum Vol 1, Learning Module 2, pp. 54–58.

SECTION 05

Present value

Present value of any future cash flow is just its FV discounted back at r — the inverse operation of compounding

Everything in DCF — bonds, stocks, projects — is a weighted sum of applications of this formula.

$$PV = FV_n / (1 + r)^n$$

where: Move money BACKWARDS on the timeline by dividing by the compounding factor.

£100 in 1y @ 5%

PV = 100 / 1.05 = £95.24

£100 in 10y @ 5%

PV = 100 / 1.05^10 = £61.39

£100 in 30y @ 5%

PV = 100 / 1.05^30 = £23.14

Source: CFA Institute, Curriculum Vol 1, Learning Module 2, pp. 54–60.

Annuities and perpetuities have closed-form present values that collapse an entire cash-flow stream into a single number

A perpetuity is simply an annuity as n → ∞ — the (1+r)^(−n) term vanishes.

PV OF ORDINARY ANNUITY

$$PV = PMT \times [1 - (1+r)^{(-n)}] / r$$

Example: £10,000/yr for 20y at 6%

$PV = 10{,}000 \times [1 - 1.06^{-20}] / 0.06$
$= 10{,}000 \times 11.4699$
$= £114{,}699$

PV OF PERPETUITY

$$PV = PMT / r$$

Example: UK consol paying £50/yr at 5%

$PV = 50 / 0.05$
$= £1{,}000$

Ordinary annuity with n → ∞ collapses to the same formula because (1+r)^(−n) → 0.

Source: CFA Institute, Curriculum Vol 1, Learning Module 2, pp. 56–60.

A bond's price is the present value of its promised coupons plus the present value of par — nothing more, nothing less

Zero-coupon bonds collapse the pricing formula down to a single term — PV of face value.

$$P = \Sigma \; C / (1 + r)^t + FV / (1 + r)^n$$

where: C = periodic coupon; FV = face value at maturity; r = per-period YTM; n = periods to maturity.

CFA LM 2 — INDIA SOVEREIGN ZERO-COUPON BOND

20-year Government of India zero-coupon bond, face value INR 100, YTM 6.7%. Compute the price.

1. C = 0 (zero-coupon). Only one cash flow: FV = INR 100 at n = 20.
2. Price = 100 / (1.067)^20
3. (1.067)^20 = 3.634
4. Price = 100 / 3.634 = INR 27.51

Price = INR 27.51 — the bond trades at roughly 27.5% of face, reflecting the 20-year discount at 6.7%.

Source: CFA Institute, Curriculum Vol 1, Learning Module 2, pp. 60–65.

A bond trading below par carries a discount that unwinds back to par at maturity — a predictable pull-to-par

Coupon < YTM ⇒price < par. Coupon > YTM ⇒price > par (premium). Coupon = YTM ⇒price = par.

CFA LM 2 — GREECE 5-YEAR SOVEREIGN BOND

5-year Greek government bond, face value EUR 100, annual coupon 3.5%, market YTM = 5.0%. Compute the price.

1. Cash flows: 3.5, 3.5, 3.5, 3.5, 103.5 (last year includes face)
2. PV of coupons = 3.5 × [1 − (1.05)^(−5)] / 0.05 = 3.5 × 4.3295 = EUR 15.15
3. PV of face = 100 / (1.05)^5 = 100 / 1.2763 = EUR 78.35
4. Price = 15.15 + 78.35 = EUR 93.49

Price = EUR 93.49. CFA text rounds to EUR 93.091; both answer under par because coupon < YTM.

PULL-TO-PAR INTUITION

Over each year of the bond's life the price drifts back toward par, independent of yield changes. The total return decomposes into a coupon income and a capital gain component that together sum to YTM if yields stay constant.

Source: CFA Institute, Curriculum Vol 1, Learning Module 2, pp. 65–70.

The Gordon-growth DDM prices a stock as the PV of a growing perpetuity of dividends — the workhorse of equity valuation

Two-stage DDM = explicit-period PV + Gordon-growth terminal. Rule: terminal growth g must be < required return r.

$$P_0 = \Sigma\ D_t / (1 + r)^t + [\ D_(N+1) / (r - g)\ ] / (1 + r)^N$$

where: Stage 1: explicit dividends D_1..D_N. Stage 2: Gordon terminal with steady growth g from N+1 onward.

CFA LM 2 — SHIPLINE INC. TWO-STAGE DDM

Current dividend D_0 = USD 2.00. High growth g_1 = 6% for 3 years. Long-run g_2 = 3% thereafter. Required return r = 10%.

1. Stage 1 dividends: $D_1 = 2.12$, $D_2 = 2.247$, $D_3 = 2.382$
2. Terminal value at t = 3: $TV = D_4 / (r - g_2) = (2.382 \times 1.03) / (0.10 - 0.03) = 2.453 / 0.07$ = USD 35.05
3. Discount cash flows: $D_1/1.10 + D_2/1.10^2 + D_3/1.10^3 + TV/1.10^3$
4. = 1.927 + 1.857 + 1.790 + 26.329 = USD 31.90

P_0 ≈ USD 31.90 — 82% of the value comes from the terminal, typical for DDMs with moderate explicit periods.

Source: CFA Institute, Curriculum Vol 1, Learning Module 2, pp. 71–76.

SECTION 06

NPV, IRR & applications

NPV tells you the dollar value of a project today; accept if positive, reject if negative — unambiguously correct under any capital structure

NPV compounds the promise: a positive NPV means the project increases firm value today by exactly that amount.

$$NPV = -CF_0 + \Sigma \; CF_t / (1 + r)^t \qquad \text{accept iff } NPV > 0$$

where: CF_0 = initial outlay (negative by convention); r = cost of capital = required return on the project.

PROJECT EVALUATION EXAMPLE

Initial outlay USD 1,000; after-tax cash flows 300, 400, 400, 300 over years 1–4. Cost of capital 10%.

1. PV(y1) = 300 / 1.10 = 272.73
2. PV(y2) = 400 / 1.10^2 = 330.58
3. PV(y3) = 400 / 1.10^3 = 300.53
4. PV(y4) = 300 / 1.10^4 = 204.90
5. NPV = −1,000 + 272.73 + 330.58 + 300.53 + 204.90 = +108.74

NPV = +USD 108.74 → ACCEPT. The project increases shareholder value by ~USD 109 today.

Source: CFA Institute, Curriculum Vol 1, Learning Module 2, pp. 76–82.

IRR is a useful shorthand but breaks down with non-conventional cash flows or mutually exclusive projects — always cross-check with NPV

IRR is the r at which NPV = 0. It's a useful comparable but the NPV rule is the theoretically correct decision rule.

FAILURE 1 · MULTIPLE IRRs

A cash-flow pattern with sign changes in both directions (−, +, −) can satisfy NPV = 0 at more than one rate.

Example: mining project with end-of-life clean-up outflow.

FAILURE 2 · SCALE MISMATCH

Project A: invest 100, earn 150 (IRR 50%). Project B: invest 1,000, earn 1,300 (IRR 30%).

IRR prefers A. NPV at 10% says B creates much more value. B wins.

FAILURE 3 · TIMING MISMATCH

Front-loaded cash flows look good under IRR (high reinvestment assumption), back-loaded under NPV (actual cost of capital).

Crossover rate = discount rate at which NPVs match. Below it, NPV and IRR disagree.

[1] IRR implicitly assumes intermediate cash flows are reinvested at the IRR — often an unrealistic assumption above the WACC.

[2] For mutually exclusive projects with different scales, NPV wins the tie every time — IRR can prefer a smaller project.

Source: CFA Institute, Curriculum Vol 1, Learning Module 2, pp. 78–82.

A level mortgage payment is just the PMT of an annuity that discounts to the loan amount — the same annuity formula solved the other way round

The loan balance is the PV of remaining payments. Early payments are mostly interest; late payments are mostly principal.

$$PMT = PV \times r / [\ 1 - (1 + r)^{(-n)}\ ]$$

where: PV = loan principal; r = per-period rate; n = number of payments (monthly over loan life).

CFA LM 2 — USD 800,000 MORTGAGE · 30-YEAR · 5.25% NOMINAL

Loan PV = 800,000. Nominal 5.25% annual → monthly rate = 5.25%/12 = 0.4375%. n = 360 months.

1. Denominator: 1 − (1.004375)^(−360) = 1 − 0.2092 = 0.7908
2. Numerator: 800,000 × 0.004375 = 3,500
3. PMT = 3,500 / 0.7908 × (1 / 0.004375) simplifies directly to USD 4,418.11 per month
4. Total paid over 30 years: 4,418.11 × 360 = USD 1,590,519 → USD 790,519 of interest on USD 800,000 principal

USD 4,418.11 monthly payment. First month: 3,500 interest / 918 principal. Last month: 19 interest / 4,399 principal.

Source: CFA Institute, Curriculum Vol 1, Learning Module 2, pp. 58–60.

Every TVM problem reduces to a single question: where on the timeline is each cash flow, and at what rate do we move them?

The six archetypes below cover every calculation in this deck. Identify the pattern first; apply the formula second.

Archetype	Formula	Typical use
Single CF — FV	$FV = PV \times (1 + r)^n$	Savings growth
Single CF — PV	$PV = FV / (1 + r)^n$	Zero-coupon bond
Ordinary annuity — PV	$PV = PMT \times [1 - (1+r)^{(-n)}] / r$	Loan amortization
Ordinary annuity — FV	$FV = PMT \times [(1+r)^n - 1] / r$	Retirement contributions
Perpetuity — PV	$PV = PMT / r$	Consols, preferred stock
Growing perpetuity — PV	$PV = PMT / (r - g)$	Gordon-growth DDM

FOUR-STEP DECISION RULE

1. Draw the timeline. 2. Identify the archetype. 3. Pick the matching formula. 4. Check units: does r match the period of n?

Source: CFA Institute, Curriculum Vol 1, Learning Module 2, Summary.

KEY TAKEAWAYS

Deck 1 — Time Value of Money & DCF Foundations

- r is simultaneously a required return, a discount rate, and an opportunity cost — decomposable into r_real-rf + inflation + default + liquidity + maturity premia.
- Time-weighted return evaluates managers (GIPS standard); money-weighted evaluates the investor's actual wealth experience.
- Gross → Net → After-tax → Real → Leveraged: always know which of these five a quoted return refers to.
- Two-stage DDM (Shipline): explicit dividends + Gordon terminal; terminal value usually dominates total PV.
- Arithmetic mean ≥ Geometric mean ≥ Harmonic mean; pick the mean that matches the question (expected next-period vs actual compound vs dollar-cost-averaging).
- EAR = (1 + r/m)^m − 1; continuous compounding is the limit with FV = PV × e^(r·n); rates are additive under continuous compounding.
- Bond price = PV of coupons + PV of face; India 20y zero at 6.7% → 27.51; Greece 5y 3.5% coupon at 5% YTM → 93.49.
- NPV rule dominates IRR: NPV > 0 ⇒accept, unambiguous under any capital structure. IRR can mislead with non-conventional CFs, scale, or timing.